

Deliberation by Design: A Multi-Agent Large-Language-Model System for Constructively Aligned, Provenance-Track Examination Authoring in the Sciences

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Abstract

Authoring a high-quality summative examination requires the simultaneous satisfaction of constraints that are individually well understood but jointly difficult to hold in mind: content coverage in the proportions a course promises, cognitive demand matched to stated learning outcomes, freedom from item-writing flaws, accessibility for students with diverse needs, and defensible measurement properties. A single author working under deadline tends to collapse these constraints into whichever is easiest to satisfy, producing exams that drift toward recall, harbour cueing flaws, and embed construct-irrelevant difficulty. We present an agentic system that reframes exam construction as *structured disagreement* rather than single-pass generation. A team of specialised large-language-model (LLM) agents — a subject-matter expert, a blueprint architect, and a panel of adversarial critics for alignment, item-writing mechanics, accessibility, psychometrics, and test-wiseness — propose, critique, and refine items across bounded epochs, coordinated by a *deterministic* (non-LLM) moderator that applies a faculty-specified trade-off policy and records every proposal, objection, and resolution as an immutable provenance event. Item proposals are grounded in the instructor’s own uploaded materials by mandatory retrieval citations and an independent grounding check, so the exam is built from course content rather than from the model’s parametric memory. We describe the design, its grounding in established assessment theory (backward design, Bloom’s revised taxonomy, constructive alignment, and evidence-centred design), and an end-to-end reference implementation that produces a complete, render-ready exam bundle — exam, answer key, rubrics, accessibility variants, a coverage and difficulty report, and a full audit trail — in roughly three minutes of wall-clock time on a two-tier model routing (a strong hosted model for generative roles and a small hosted model for verification roles); the same pipeline can be run entirely on local open-weight models at the cost of speed. We discuss the pedagogical, cost, and time trade-offs of the approach, including its limitations (the absence of empirical item-response data, the residual risk of fluent-but-wrong content, and the gap between estimated and observed psychometrics), and we describe our plan to release the system as an open-source project to invite scrutiny, replication, and disciplinary refinement.

1 Introduction

Summative assessment is among the highest-stakes design tasks an instructor performs, and among the least resourced. A midterm or final examination is the instrument through which a course converts months of instruction into defensible inferences about what students know and can do. Yet the typical author of that instrument is a single faculty member, working under deadline, holding in mind at once a set of constraints that the educational-measurement literature treats as

distinct specialities: the exam must cover the right topics in the right proportions; it must target the cognitive levels the course outcomes actually promise rather than the levels that are easiest to write items for; it must be free of the grammatical cueing, implausible distractors, and overlapping options that the item-writing literature has catalogued for decades (Haladyna et al. 2002); it must remain accessible to students with diverse needs and language backgrounds, rather than penalising them through construct-irrelevant difficulty (Messick 1995); and it must distinguish students who have understood the material from those who have not, within a fixed time and point budget.

Each of these dimensions has a mature literature behind it. Backward design (Wiggins and McTighe 2005) tells us to specify the evidence an assessment must elicit *before* writing any tasks. Bloom’s revised taxonomy (Anderson and Krathwohl 2001) gives a vocabulary for cognitive demand. Constructive alignment (Biggs 1996) requires that each task elicit evidence at the cognitive level its target outcome promises. Evidence-centred assessment design (Mislevy et al. 2003) frames every item as an argument linking a claim about the student to the evidence a task elicits. The *Standards for Educational and Psychological Testing* (American Educational Research Association et al. 2014) codify validity and fairness as professional obligations. Few practising faculty, however, have the time to consult these literatures for each assessment they write, and the predictable consequence is a well-documented set of failure modes: silent drift toward recall and procedural application, item-writing flaws that inflate scores without measuring anything, and accommodations produced reactively as ad hoc edits to a finished exam rather than designed in from the start.

The recent maturation of large language models (LLMs) (Brown et al. 2020) has prompted enthusiasm for automating assessment authoring, and a substantial body of work on automatic question generation predates the current generation of models (Kurdi et al. 2020). The naïve application of a single, powerful model to “write me an exam,” however, reproduces the single-author failure mode in a new guise. Asked to optimise content, cognition, accessibility, and measurement simultaneously, a single model tends to produce a smooth compromise that *looks* plausible on every dimension while obscuring where the trade-offs were made — an argument we make on design grounds rather than from a controlled comparison. Worse, LLMs are prone to confident fabrication (Ji et al. 2023) and to sycophantic agreement that simulates deliberation without its substance (Sharma et al. 2023; Perez et al. 2023). A generator that hallucinates a plausible-but-wrong answer key, or that agrees with every critique it receives, is not merely unhelpful; it is dangerous in a context where the output drives grades and is subject to appeal.

This paper presents a system designed around the opposite premise. We treat exam construction not as a generation problem but as a *deliberation* problem, and we make the deliberation structural. A small team of specialised LLM agents — a subject-matter expert (SME), a blueprint architect, and a panel of adversarial critics — propose and contest items across bounded epochs. A **deterministic** moderator, not itself an LLM, collects objections, applies a trade-off policy the faculty member specifies in advance, and records every move as an immutable provenance event. Item content is **grounded** in the instructor’s own uploaded materials through mandatory retrieval citations and an independent grounding check, so the exam is assembled from course content rather than from the model’s parametric memory. The instructor of record is retained as the authoritative decision-maker, reviewing a fully rendered bundle rather than mediating the machinery.

Our contributions are: (1) a system design that operationalises four established assessment frameworks as hard constraints in an agentic pipeline, rather than as advisory prompts; (2) a deliberately *non-LLM* moderator and a mandatory-objection protocol that together counter the multi-agent failure mode of convergence-through-agreement; (3) a grounding regime that ties every item to cited source passages and audits answer-key support; (4) a full-provenance record engineered for

defensibility against grade appeals; and (5) a working, end-to-end reference implementation, which we describe and which we are preparing to release as open source. We are explicit throughout about what the system does *not* yet do — most importantly, it produces *estimated* rather than empirically observed psychometrics — and we frame the open-source release as the mechanism by which those gaps are most credibly closed.

2 Related Work

Automatic item and question generation. Automatic question generation (AQG) has a long pre-LLM history, comprehensively surveyed by Kurdi et al. (2020), who reviewed 93 studies and noted that the field had paid little attention to controlling difficulty, enriching question structure, or generating feedback, precisely the dimensions a blueprint-driven, critic-mediated design foregrounds. Classical AQG systems were largely template- or syntax-driven and produced factual recall questions; they rarely engaged the cognitive-alignment or fairness concerns that dominate professional assessment practice. The arrival of instruction-tuned LLMs (Brown et al. 2020) dramatically expanded the fluency and range of generated items, and recent work has applied LLMs to question generation and to Bloom-level-targeted item writing. Our work differs in treating generation as only the first move in a multi-party deliberation, and in subordinating the generator to an explicit blueprint and a panel of critics.

Multi-agent LLM systems. A growing literature shows that orchestrating several LLM instances — debating, critiquing, or playing distinct roles — improves factuality and reasoning relative to a single pass. Du et al. (2024) demonstrate that multi-agent debate reduces hallucination and improves reasoning; Park et al. (2023) show that role-specialised agents can sustain coherent, differentiated behaviour; and frameworks such as AutoGen (Wu et al. 2023) provide general machinery for multi-agent conversation. Our design draws on this line of work but departs from it in two deliberate ways. First, the coordinating moderator is *deterministic* rather than a model: the trade-off resolution that decides which dimension wins a conflict is a priority-ranked policy the faculty member sets, not an emergent property of a conversation. We made this choice because the resolution of a content-versus- accessibility conflict is a *value judgment* that must be auditable and must belong to the instructor, not to an opaque negotiation. Second, we adopt an anti-sycophancy protocol (critics must produce objections at a configurable minimum rate in early epochs, and empty critique is rejected and re-prompted) to counter the well-documented tendency of LLMs toward agreeable convergence (Sharma et al. 2023; Perez et al. 2023).

Grounding and retrieval. Retrieval-augmented generation (Lewis et al. 2020) conditions model output on retrieved documents to improve factual accuracy and reduce hallucination. We apply this idea as a *gating constraint* rather than a quality nudge: an item proposal that does not cite at least one retrieved chunk from the instructor’s materials is rejected before it can enter the item bank, and an independent grounding verifier checks that the cited passages actually support the claimed answer. Prior examinations, when supplied, are indexed for *style* only and are never returned as a content source for new items.

Assessment theory. The system is an attempt to make four established frameworks operative rather than aspirational. Backward design (Wiggins and McTighe 2005) is enforced by forbidding item generation before a blueprint is approved. Bloom’s revised taxonomy (Anderson and Krathwohl 2001) is the vertical axis of that blueprint. Constructive alignment (Biggs 1996) is checked per item by a dedicated agent. Evidence-centred design (Mislevy et al. 2003), together with modern validity theory (Messick 1995; Kane 2013), motivates the provenance record that turns each item

into a documented argument. Accessibility is treated as a first-class design constraint in the spirit of Universal Design for Learning (Rose and Meyer 2002; CAST 2018) rather than as post-hoc accommodation.

3 System Design

3.1 Overview and design commitments

The system is delivered as a web application but its substance is a team of agents that propose, critique, and refine items under instructor supervision. Five commitments shape the design and distinguish it from a prompt-engineered single-model generator.

1. **Multi-agent, not single-agent.** Each critic — alignment, item-writing, accessibility, psychometrics, adversarial student — is a separate agent with its own persona and output schema. The point is to make trade-offs *visible*, not to optimise them away.
2. **The moderator is deterministic.** Disagreement is resolved by a priority-ranked policy, not by a model. This keeps value judgments with the instructor and keeps resolution auditable.
3. **Grounding is mandatory.** Every item proposal must cite at least one source chunk from uploaded materials, and a grounding verifier checks that the cited chunks support the answer. Ungrounded items are auto-rejected. Prior exams are indexed for style only.
4. **Full provenance.** Every proposal, edit, objection, and resolution writes a structured provenance event. This is what makes an exam defensible under appeal.
5. **No agent-orchestration frameworks.** All agents and orchestration logic are written directly against provider SDKs with an explicit, inspectable state machine, so that the control flow — particularly the deterministic moderator — is fully legible and testable.

3.2 Inputs

The instructor provides three things. **Materials** are course readings, lecture notes, problem sets, and lab manuals (PDF and related formats); optional prior exams are accepted for style reference only. A **course specification** lists course learning outcomes (CLOs), each tagged with a target Bloom level, a weighted topic list, and free-text guiding principles (e.g., “emphasise quantitative reasoning; no rote definitions”). An **exam specification** fixes the exam type, total points and time budget, item-type counts, target difficulty distribution, and any required accessibility variants. A fourth input deserves emphasis: the **trade-off policy**, a ranked priority over *content fidelity*, *cognitive alignment*, *accessibility*, *discrimination*, and *brevity*, together with a maximum epoch count and a convergence rule. Requiring the instructor to rank these in advance places the responsibility for value judgments where it belongs and prevents the system from silently defaulting to whichever dimension is easiest to quantify.

3.3 Agent roster

Table 1 lists the agents. Each is defined by a system prompt (its “constitution”), a set of allowed actions, and a forced-JSON output schema validated against a Pydantic contract.

Table 1: The agent roster. The moderator is the only non-LLM participant.

Agent	Role	Tier
Subject-Matter Expert (SME)	Extracts themes; proposes grounded items; defends or revises under critique	High
Blueprint Architect	Builds the topic $\times$ Bloom matrix governing coverage	High
Learning-Outcomes Alignment (LOA)	Verifies each item targets the cognitive level its CLO promises	Low
Item-Writing Specialist (IWS)	Catches cueing, implausible distractors, and other mechanical flaws	Low
Accessibility Expert	Reviews readability and construct-irrelevant difficulty; builds variants	Low
Psychometrician	Estimates difficulty and discrimination; audits the distribution	Low
Adversarial Student	Attempts each item using only test-wiseness, no materials	High
Grounding Verifier	Checks cited chunks actually support the claimed answer	Low
Moderator	Deterministic orchestrator; applies the trade-off policy	— (not an LLM)

The “tier” column reflects a deliberate cost decision discussed in Section 5.2: generative, reasoning-heavy roles (SME, architect, adversarial student) are routed to a strong hosted model, while verification and rule-application roles (alignment, item-writing, accessibility, psychometrics, grounding) are routed to a small, inexpensive model.

3.4 Orchestration: the epoch loop

The pipeline runs in four phases, gated in the original design by three faculty checkpoints (CP1: blueprint approval; CP2: item-bank review; CP3: final review).

- **Phase 0 — Intake.** Materials are ingested, chunked with a heading-aware chunker, embedded, and stored in a local vector index. The specifications and trade-off policy are parsed.
- **Phase 1 — Blueprinting.** The SME extracts and ranks candidate themes from the materials; the Blueprint Architect builds a topic $\times$ Bloom matrix with target item counts and points per cell. The blueprint is the central artifact, and it is frozen at CP1 before any item is written — the structural enforcement of backward design.
- **Phase 2 — Item generation.** For each blueprint cell, the SME *over-generates* candidate items (each carrying mandatory source citations), the IWS performs a mechanics cleanup pass, and the LOA verifies constructive alignment, dropping misaligned items. When the LOA rejects an item for a Bloom mismatch, the system first attempts an automatic realignment

(re-mapping the item’s Bloom level or CLO reference and re-verifying) before discarding it.

- **Phase 3 — Refinement epochs.** In each epoch, the critic panel reviews every surviving item *in parallel*: the Accessibility Expert flags readability and construct-irrelevant difficulty, the Adversarial Student attempts each item using only test-wiseness, and the Psychometrician scores difficulty and flags distributional imbalance. The moderator collects the objections, routes each to the SME for an edit or a rebuttal, applies the trade-off policy when critics conflict, and re-verifies any edited item through IWS $\rightarrow$ LOA $\rightarrow$ grounding. The loop exits when no critical- or high-severity objections are raised for a full epoch, or when the maximum epoch count is reached.
- **Phase 4 — Finalisation.** The Accessibility Expert generates the required accommodation variants; the Psychometrician produces an exam-level report (Bloom distribution, estimated difficulty curve, CLO-coverage heatmap); and the system renders the export bundle.

3.5 The deterministic moderator

The moderator does the coordinating, and it is the one component we deliberately kept *out* of the language-model layer. For each open objection it applies a fixed procedure: a critical-severity objection forces an SME edit or item rejection; a conflict between objections on the same item is resolved by consulting the instructor’s priority ranking and selecting the higher-ranked dimension, with the rationale logged; everything else is routed to the SME for an accept-edit, rebut, or defer response. Rebut–reassert cycles are bounded before escalation. Every resolution writes a provenance event. Because this logic is deterministic, it is unit-testable, reproducible, and legible to a faculty member who wants to know exactly why a given item survived or died — properties that an LLM-mediated negotiation cannot offer.

3.6 Anti-sycophancy by construction

Multi-agent LLM systems are prone to convergence through mutual agreement, a failure mode that produces the appearance of deliberation without its substance (Sharma et al. 2023; Perez et al. 2023). We counter this structurally. Critic personas are written to reward substantive disagreement; the moderator’s mandatory-objection protocol rejects an empty early-epoch critique and re-prompts; and the Adversarial Student operates only on item text, with no access to materials or answer key, so its successes are genuine test-wiseness exploits rather than informed solving. We note only an informal observation from the handful of reference runs we have inspected so far — rebuttals are frequent and substantive rather than perfunctory — and we flag it as anecdotal, not a measured rate; quantifying objection and rebuttal behaviour across many runs is future work.

3.7 Grounding and provenance

Every SME item proposal must cite at least one retrieved chunk; proposals without citations are auto-rejected, and a separate Grounding Verifier confirms that the cited chunks support the claimed answer, routing failures back to the SME. This is retrieval-augmented generation (Lewis et al. 2020) used as a *gate* rather than a hint. The provenance record — which agent proposed each item, which objections were raised, how each was resolved, and against which source passages — is the artifact that operationalises evidence-centred design (Mislevy et al. 2003) and modern validity theory (Kane 2013): it turns each item into a documented argument and gives an instructor a concrete, auditable record to put before a grade-appeal committee.

4 Reference Implementation

We built a complete, end-to-end implementation to validate the design rather than to ship a product. It comprises roughly thirty Pydantic data contracts, one agent class per persona (each loading its constitution from an editable Markdown file), a provider abstraction supporting a hosted model, an OpenAI-compatible institutional endpoint, and a local model server, a retrieval layer over a local vector store with a local embedding model, an append-only event log with per-call input/output sidecars, the deterministic moderator state machine, and an export module that renders the final bundle through Quarto. A five-page observation UI lets a faculty member launch a run, watch an interactive timeline of every agent call, inspect outputs, edit personas, and read the documentation.

Two implementation realities are worth reporting because they generalise to anyone building agentic pipelines on current models. First, **model output is not reliably schema-conformant**. Models stringify nested JSON fields, emit invalid escape sequences when embedding LaTeX backslashes, and omit required fields when terse. We addressed this principally with a retry-on-validation loop — when a response fails its Pydantic contract, the validation error is fed back to the model and the call is retried — backed by defensive defaults and per-cell/per-item safe wrappers so that one malformed response can never collapse a parallel fan-out. Second, **concurrency must be governed per backend**. A single-GPU local model serialises internally and times out under naïve parallelism; a free institutional endpoint has an hourly quota. Per-provider semaphores (and a raised token floor for local “thinking” models that emit long hidden reasoning streams before producing content) were necessary for the parallel critique phase to complete. We report these not as incidental engineering but as evidence that robustness scaffolding is a first-class concern in agentic assessment tooling.

On a two-tier routing — a strong hosted model for the high tier and a small hosted model for the low tier — a quiz-scale reference corpus runs the full Phase 0–4 pipeline, including four refinement epochs, in roughly three minutes of wall-clock time, and produces a complete bundle: the exam, answer key, instructor notes, rubrics, an exam report, and a machine-readable provenance file, in PDF, DOCX, LaTeX, and HTML, together with an on-demand human-readable narrative of the run. We emphasise that this is a *functional* demonstration on a small corpus, not a controlled study of exam quality; the distinction is central to the limitations we discuss below.

5 Discussion: Trade-offs of the Approach

We assess the approach along the dimensions that matter to a department deciding whether to adopt it, and we try not to flinch at the unfavourable ones.

5.1 Pedagogical trade-offs

The strongest pedagogical argument for the system is that it makes the *deliberative structure* of good assessment design unavoidable. Backward design, constructive alignment, Bloom-level targeting, and item-writing discipline are not suggestions layered on top of a generator; they are enforced by the control flow. A faculty member cannot get items before approving a blueprint, cannot keep a misaligned item that the LOA rejects without an explicit override, and cannot ship an item whose answer is unsupported by cited materials. This is the educational-measurement equivalent of code review, and it is aimed squarely at the kinds of failure modes the literature warns about (Haladyna et al. 2002; Messick 1995), though whether it reduces them in practice is an empirical question we have not yet answered. Treating accessibility as a first-class critic, rather than a post-hoc accommodation, aligns the workflow with Universal Design for Learning (Rose and Meyer 2002).

The counter-argument is that the system can *displace* rather than *develop* faculty expertise. A tool that silently fixes cueing flaws does not teach the instructor to avoid them next time, and an instructor who defers to the machine’s blueprint may exercise less of the curricular judgment the blueprint is meant to externalise. We have tried to mitigate this by making the system observational and the provenance legible — the faculty member sees *why* each decision was made — but the risk of deskilling is real and is best evaluated empirically in a pilot. There is also a genuine pedagogical hazard specific to LLMs: a fluent but subtly wrong item or answer key (Ji et al. 2023) is more dangerous than an obviously wrong one, because it survives a casual read. Mandatory grounding and the adversarial critic reduce this risk but do not eliminate it, which is why the instructor checkpoint on the rendered bundle is not optional in the intended workflow.

5.2 Cost trade-offs

Cost is governed almost entirely by model routing. The two-tier scheme is the key economic lever: a typical run issues on the order of a few dozen high-tier (generative) calls and roughly one hundred low-tier (verification) calls. Routing the high tier to a strong hosted model and the low tier to a small or local model concentrates spend on the generative roles whose output most shapes the exam, while keeping the marginal cost of an exam low. We have not benchmarked per-exam cost directly; an order-of-magnitude estimate from the call counts and current hosted-model token prices is single-digit cents to a few dollars per exam, dominated by the high-tier calls. Marginal cost approaches zero when the low tier (and optionally the high tier) is served by a local model on departmental hardware, at the cost of speed and, plausibly, some output quality. The system is provider-agnostic for exactly this reason, and the local-only configuration buys an additional benefit beyond cost: full data locality, since no course materials leave the institution. The principal *non-monetary* cost is engineering and maintenance: agentic pipelines on current models require the robustness scaffolding described above, and that scaffolding must track model and SDK changes.

5.3 Time trade-offs

The time argument is the most concrete. The system reallocates faculty time from the mechanical aspects of construction — drafting items, checking distractors, balancing coverage, building accommodation variants, assembling an answer key and rubrics — to the consequential ones: specifying outcomes, ranking trade-offs, and reviewing a finished draft. The marginal machine time to produce a full bundle is minutes. The faculty time is front-loaded into the course and exam specifications, which, for a recurring course with stable outcomes, amortises across semesters: the specification is written once and reused, and only the materials and exam parameters change. The break-even is least favourable for a one-off exam in a course whose outcomes have never been formalised, and most favourable for a recurring exam in a course with a stable, documented outcome set — precisely the qualifying-exam and large-enrolment-core settings where defensibility matters most.

5.4 Why multiple agents, empirically

A reasonable skeptic asks whether the multi-agent structure earns its complexity. Our position, consistent with the multi-agent-debate literature (Du et al. 2024), is that separating the critics is what makes the trade-offs *visible and contestable*. When the Accessibility Expert objects that a stem carries construct-irrelevant difficulty and the SME rebuts that the vocabulary in question *is* the construct, the disagreement is recorded, the policy is consulted, and the resolution is logged with rationale. A single agent asked to balance the same concerns internally produces a compromise

with no visible seam and no audit trail. The cost is more calls and more orchestration; the benefit is accountability, which is what a graded, appealable context demands.

6 Limitations

The boundaries of the present work are worth stating plainly.

Estimated, not observed, psychometrics. The Psychometrician’s difficulty and discrimination figures are *model estimates*, not statistics derived from student responses. Real item parameters are knowable only after administration. The system’s psychometric report should be read as a design-time sanity check on the *intended* distribution, not as a substitute for post-hoc item analysis. Closing this gap requires a feedback loop in which administered-item statistics are returned to the system — work we have not yet done.

Residual fabrication risk. Grounding and adversarial critique reduce but do not eliminate the chance of a fluent, plausible, and wrong item or answer key (Ji et al. 2023). The mandatory instructor review of the rendered bundle is the backstop, and the workflow depends on it: the system is decision-support, not an autonomous authority.

No controlled evaluation yet. The implementation is verified end-to-end on a small reference corpus. We have *not* run a controlled study comparing system-assisted exams to instructor practice, nor measured inter-rater agreement between the agent critics and human assessment experts, nor validated the constructive-alignment judgments against expert raters. These are the studies that would establish the system’s value, and they are the central items of future work below. Claims of quality improvement in this paper are therefore *architectural and theoretical*, not empirical.

Scope. The approach suits exams whose value lies in structured coverage of stable outcomes — recurring undergraduate exams, graduate qualifying exams. It is ill-suited to assessments whose pedagogical value is unpredictability (oral exams, open-ended research proposals), and we make no claim for those settings.

Generalisation of the corpus. The reference runs use a small science corpus. The theme-extraction batching path that would be exercised by a full textbook is implemented but not yet stress-tested at that scale, and discipline-specific conventions beyond the sciences (e.g., the assessment norms of the humanities) are untested.

Human-in-the-loop posture. The original design specified faculty checkpoints at CP1, CP2, and CP3. The reference implementation auto-advances these and places the human review at the rendered-bundle stage instead. This is a defensible UX choice for an observation-first tool, but it weakens the in-loop control the design envisioned; making the mid-run checkpoints optionally blocking is straightforward and is on the roadmap.

7 Future Work

Several directions follow directly from the limitations.

1. **Empirical validation.** The priority is a pilot: a small number of faculty, one course and exam each, with a structured debrief comparing the system-assisted exam to prior practice on time, item quality, and faculty experience. In parallel, an expert-rater study would measure

agreement between the agent critics (especially the LOA’s alignment judgments) and human assessment specialists.

2. **Closing the psychometric loop.** Returning administered-item statistics — difficulty, discrimination, distractor analysis — to the system would let it calibrate its estimates and would convert the design-time report into a genuine measurement instrument over repeated administrations.
3. **Programmatic anti-sycophancy enforcement and policy-driven conflict resolution.** The mandatory-objection rate and the priority-rank consultation for directly conflicting critic demands are specified in the design and only partially enforced in code; completing them tightens the deliberation guarantees.
4. **Standards interoperability and persistence.** Export to QTI / Common Cartridge for direct LMS import, and a durable persistence layer for cross-session resume and longitudinal item banking, would move the system from a single-session tool toward a departmental asset.
5. **Broader disciplinary and accessibility coverage.** Stress-testing theme extraction on full textbooks, validating non-science disciplines, and expanding the accessibility variants beyond the current set are all tractable extensions.

8 Open-Source Release and Collaboration

We intend to release the system as an open-source project, and the decision is methodological rather than merely generous. Three of the system’s commitments — auditability, a deterministic and inspectable moderator, and reproducibility — are credible only if the code is open to inspection. An assessment tool that asks a department to trust its alignment judgments, while hiding how those judgments are made, contradicts its own provenance argument. The deliberate avoidance of heavyweight agent-orchestration frameworks in favour of a plain, legible state machine was made partly with this in mind: a reviewer can read the moderator and verify that the trade-off resolution does what the policy says, without untangling a framework’s control flow.

The release is also the most credible path to the empirical validation the limitations demand. The studies that would establish the system’s worth — pilots across courses and disciplines, expert-rater agreement, psychometric calibration against administered data — are larger than any single group can run, and they depend on instrument transparency to be comparable across sites. An open project with editable agent personas (each is a Markdown file), provider-agnostic model routing, and a documented data-contract layer invites exactly the multi-site, multi-discipline contribution that turns a reference implementation into a validated method. We plan to publish the repository with the reference corpus, the agent constitutions, the deterministic moderator, and the rendering pipeline; to document the contribution surfaces (new critic agents, new export targets, new disciplinary persona sets); and to license it for unrestricted academic use. We invite the assessment and educational-technology communities to test the alignment judgments against their own expertise, to contribute critics for concerns we have not modelled, and to report where the system fails — which, for a tool whose entire premise is structured disagreement, is the most useful contribution of all.

9 Conclusion

We have described a multi-agent LLM system that treats examination authoring as structured deliberation rather than single-pass generation. Four assessment frameworks — backward design, Bloom’s revised taxonomy, constructive alignment, and evidence-centred design — enter the system

as hard constraints rather than advice. A deterministic, auditable moderator coordinates the critic agents under a trade-off policy the instructor sets. Every item is grounded in cited course materials and carries a full provenance trail. Taken together, these make the trade-offs of good assessment design visible, contestable, and defensible. A working reference implementation produces a complete, render-ready exam bundle in minutes at low marginal cost, with the instructor retained as the authoritative reviewer. The approach’s pedagogical, cost, and time trade-offs are favourable for recurring, defensibility-sensitive assessments and unfavourable for one-off or open-ended ones. Its central limitation — estimated rather than observed psychometrics, and the absence of controlled evaluation — is real, and we have argued that the most credible route to closing it is an open-source release that invites the multi-site, multi-discipline scrutiny on which assessment validity ultimately depends.

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